Kembangkan aplikasi *WebGIS full-stack* dengan fitur autentikasi dan CRUD untuk mengelola data fasilitas publik

Tugas Praktikum 9 - Pengembangan web aplikasi *WebGIS*

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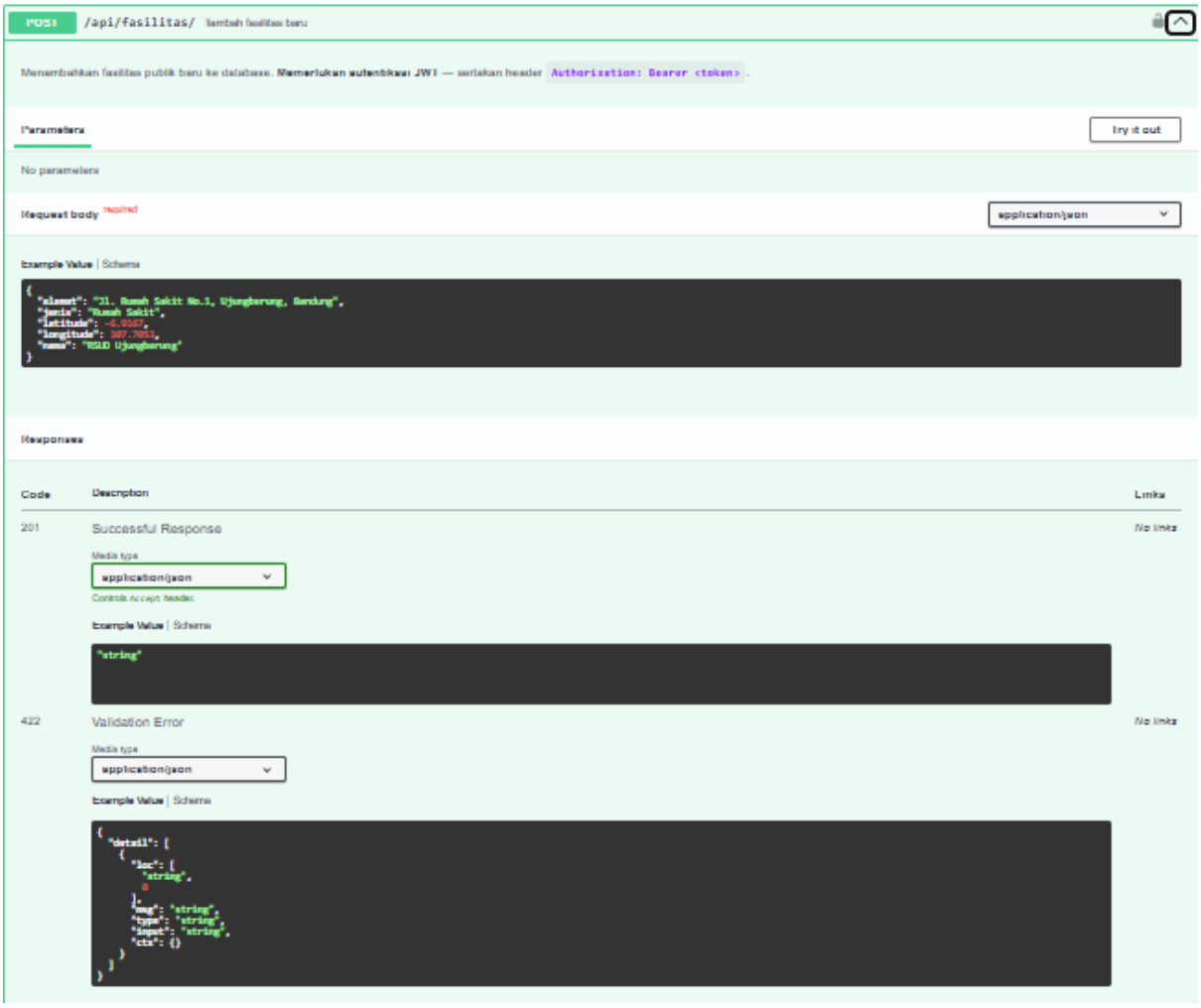
1. Implementasi CRUD

   CRUD dapat di akses melalui http://localhost:8000/docs

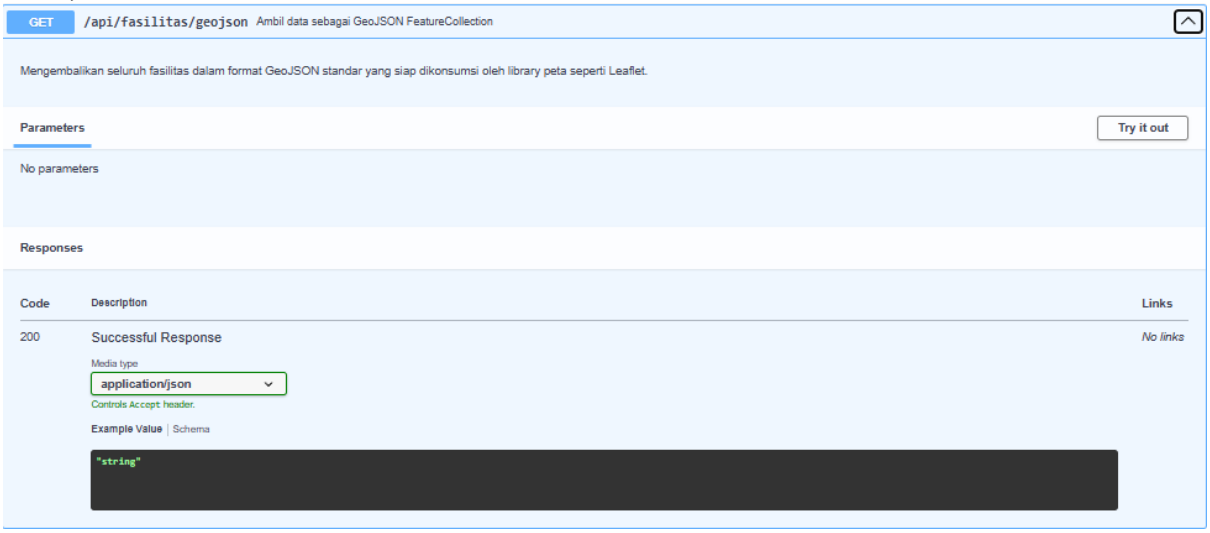
   Create:

   Ketika user mengirimkan data baru untuk membuat fasilitas, **API/Fasilitas (post)**

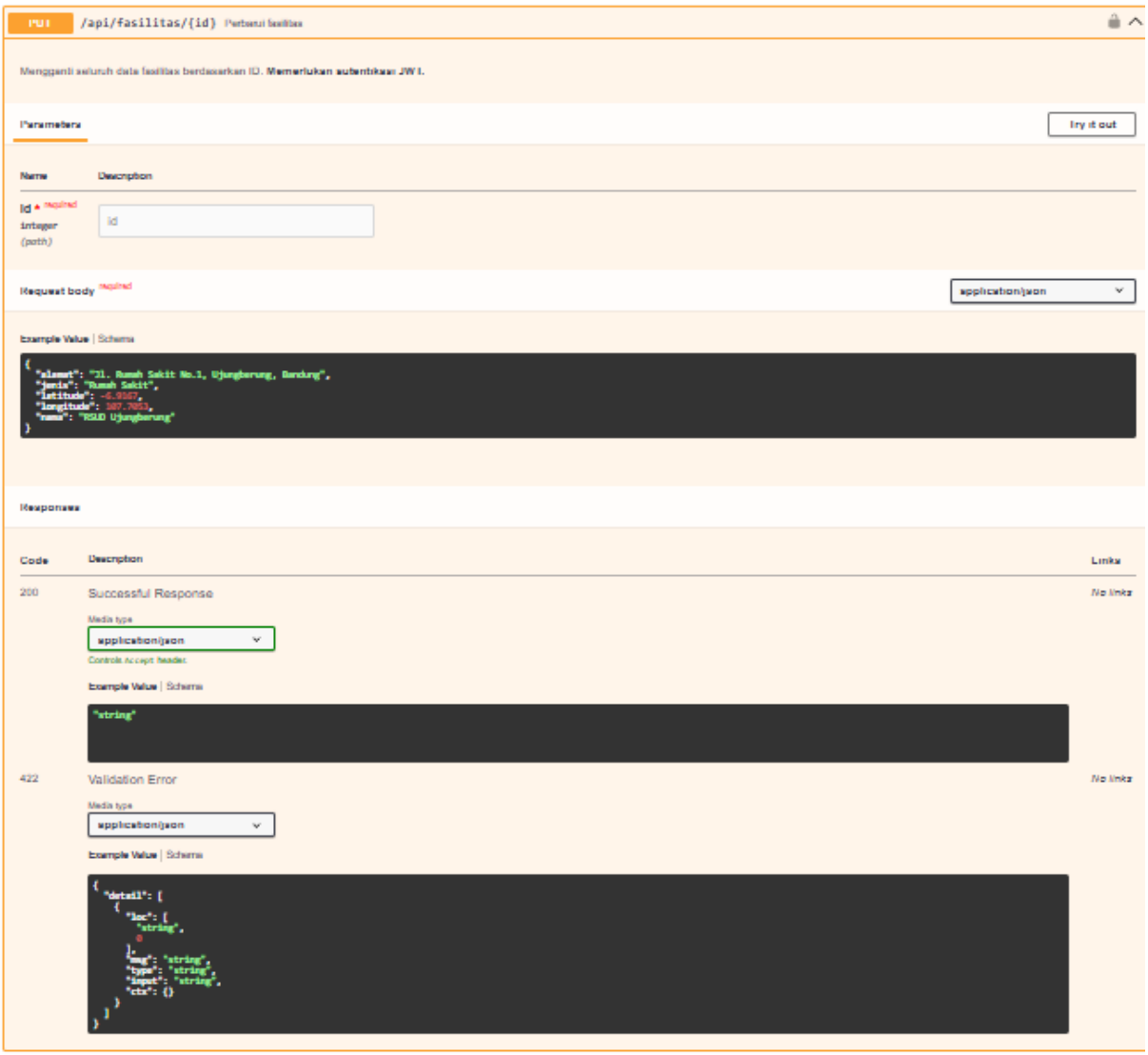


READ:

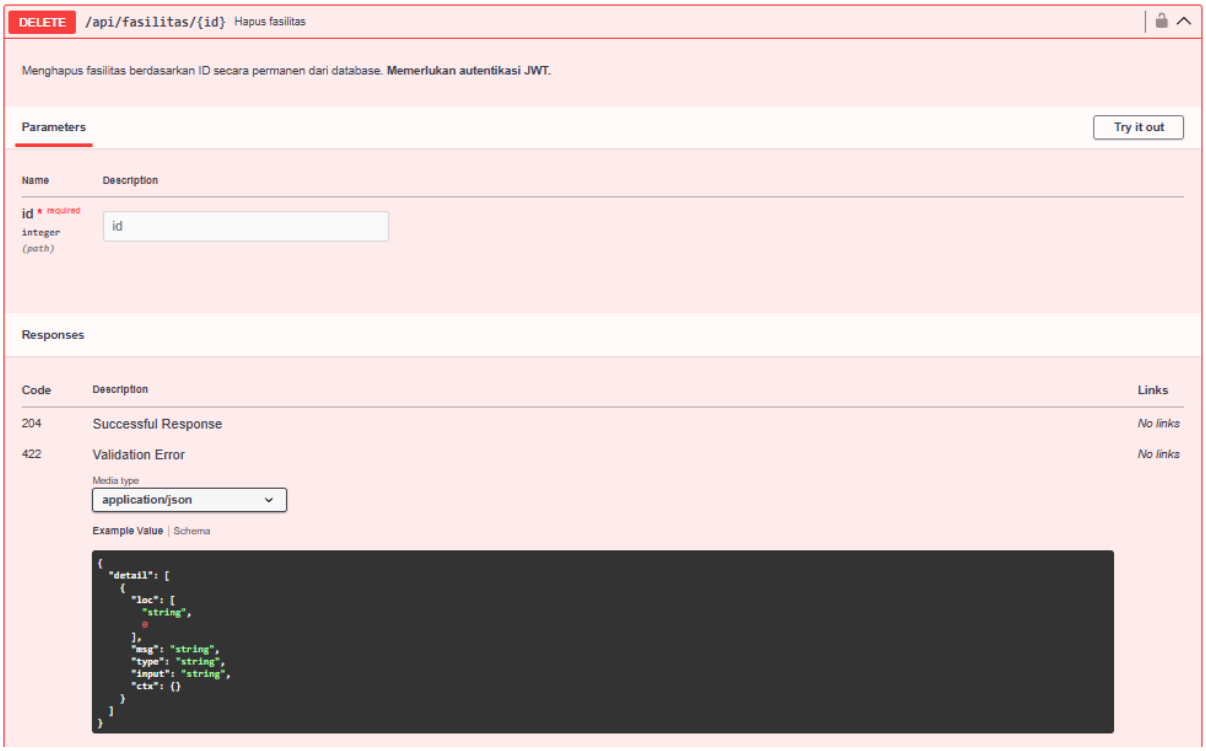
bertugas mengambil seluruh data dari database dan memformatnya secara spesifik menjadi standar ***GeoJSON FeatureCollection*** , **(/api/fasilitas/geojson - GET)**



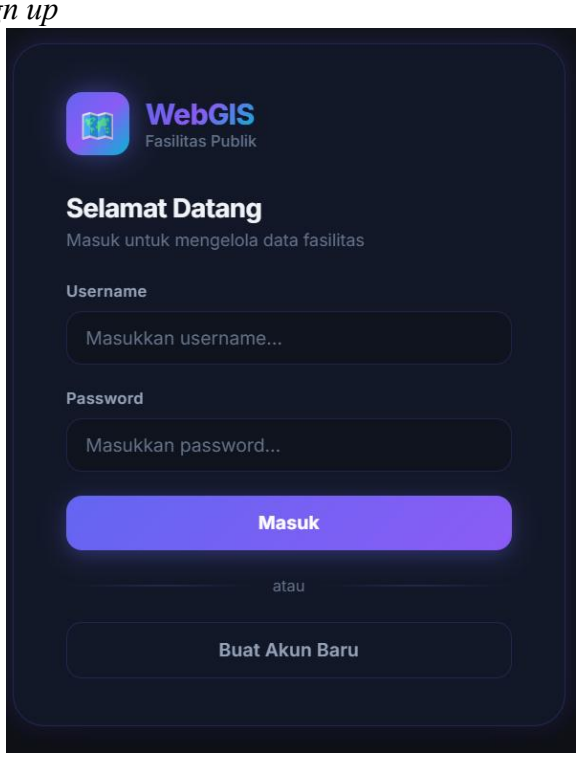
UPDATE:
Menerima request perubahan data berdasarkan parameter id. **/api/fasilitas/{id}**



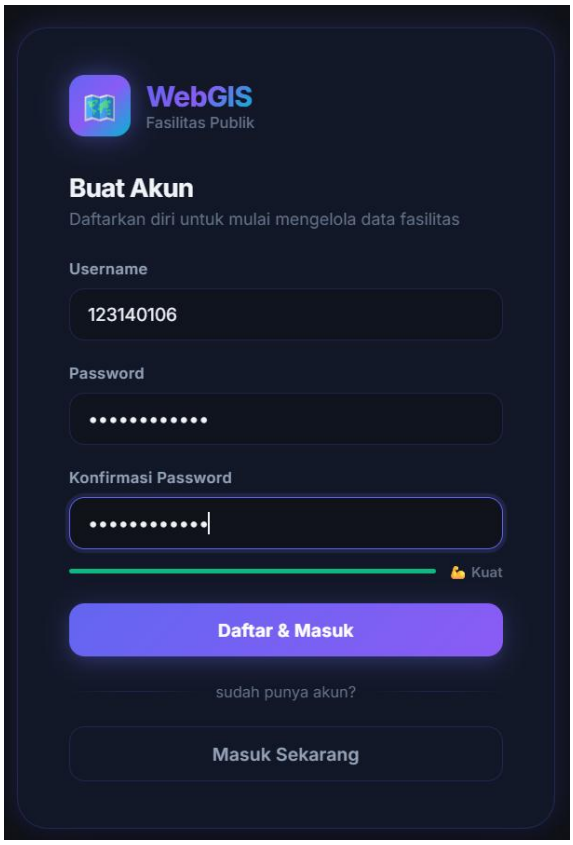
Delete:
Menghapus satu fasilitas spesifik dari database menggunakan query DELETE FROM fasilitas WHERE id=$1, **/api/fasilitas/{id}**



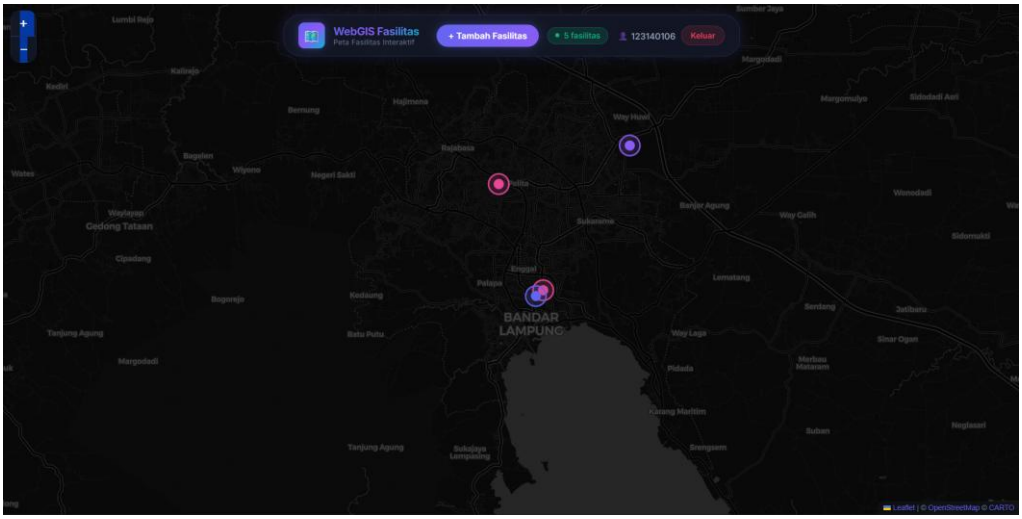
2. Implementasi login dengan JWT
   Tampilan *Log in & sign up*



   Tampilan sign up:



   Tampilan sudah didalam:



3. Form untuk add/edit fasilitas dengan validasi
   a. Validasi: untuk memvalidasi (pengecekan input/output untuk dikirimkannya ke sistem backend)

```
function validateForm(form) {
  const errors = {}
  if (!form.nama.trim()) errors.nama = 'Nama wajib diisi'
  else if (form.nama.trim().length < 3) errors.nama = 'Nama minimal 3 karakter'
  if (!form.jenis) errors.jenis = 'Jenis fasilitas wajib dipilih'
  if (form.latitude === '' || form.latitude === undefined) {
    errors.latitude = 'Latitude wajib diisi'
  } else if (isNaN(Number(form.latitude)) || Number(form.latitude) < -90 || Number(form.latitude) > 90) {
    errors.latitude = 'Latitude harus antara -90 dan 90'
  }
  if (form.longitude === '' || form.longitude === undefined) {
    errors.longitude = 'Longitude wajib diisi'
  } else if (isNaN(Number(form.longitude)) || Number(form.longitude) < -180 || Number(form.longitude) > 180) {
    errors.longitude = 'Longitude harus antara -180 dan 180'
  }
  return errors
}

```

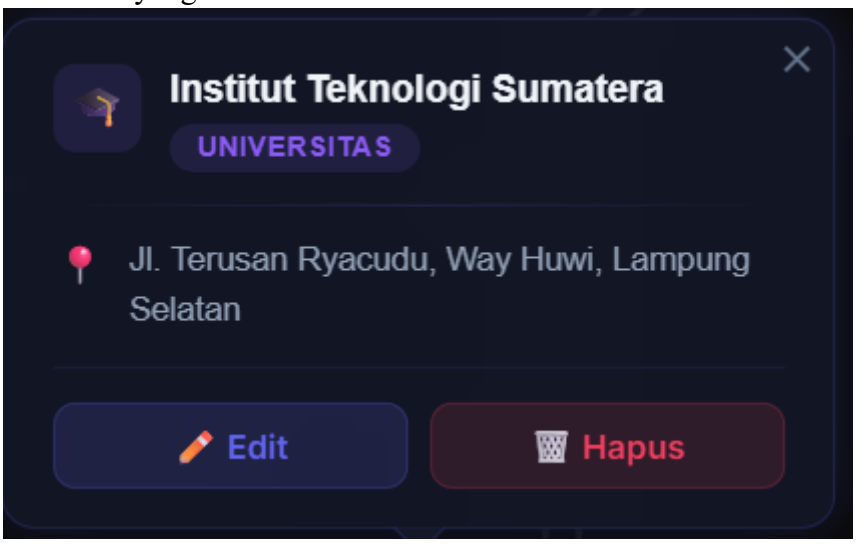
   b. Bagian "Add/Edit": unutk bagian penambahan atau melakukan pengubahan

```
const isEdit = !!(initialData && initialData.id)
```

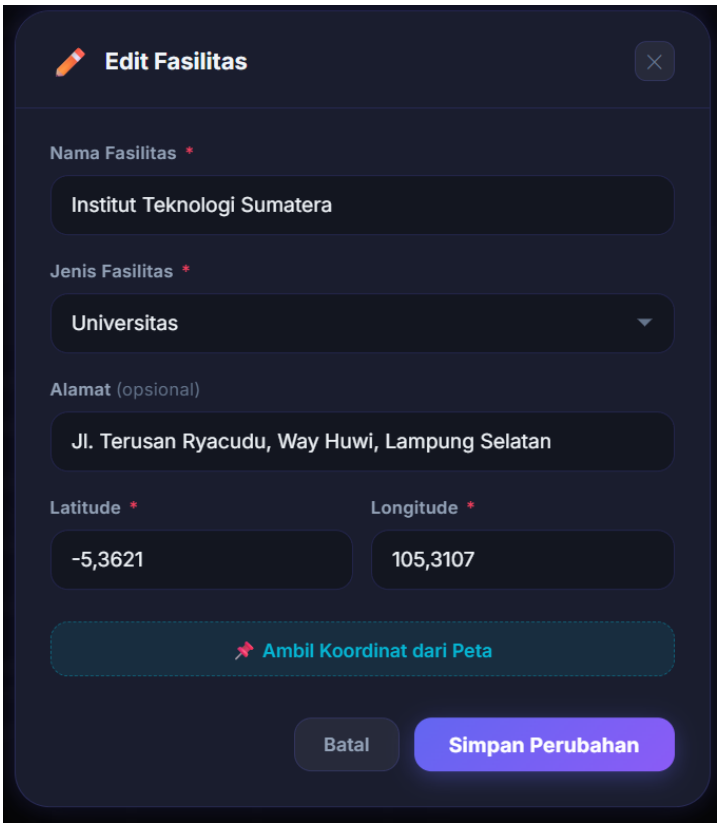
Fungsi handleSubmit:

```
{/* Modal */}
<div className="modal-card" id="fasilitas-modal" role="dialog" aria-modal="true"
     aria-label={isEdit ? 'Edit Fasilitas' : 'Tambah Fasilitas'}>

  {/* Header */}
  <div className="modal-header">
    <div className="modal-title-wrap">
      <span className="modal-icon">{isEdit ? '✏️' : '📍'}</span>
      <h3 className="modal-title">{isEdit ? 'Edit Fasilitas' : 'Tambah Fasilitas'}</h3>
    </div>
    <button id="modal-close" className="modal-close" onClick={onClose} aligned="Tutup modal">✕</button>
  </div>

  {/* Body */}
  <div className="modal-body">
    {success && (
      <div className="auth-alert auth-alert-success">
        ✅ Data berhasil {isEdit ? 'diperbarui' : 'ditambahkan'}!
      </div>
    )}
```

4. Menampilkan marker yang bisa diklik untuk edit/delete:



Edit:



Delete:

Github: https://github.com/Scorlines/SIG_106